

2024.

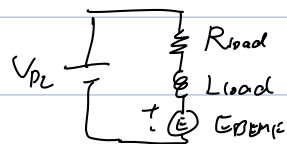
Q1. (a). $t = \frac{1}{F_{PWM}} = \frac{1}{2000} = 500 \mu s$

$t_{on} = t \cdot \text{duty cycle} = 500 \times 65\% = 325 \mu s$

$t_{off} = t - t_{on} = 175 \mu s$

$E_{BEMF} = 0.5 \times \frac{6000}{1000} = 2 \checkmark$

on state:

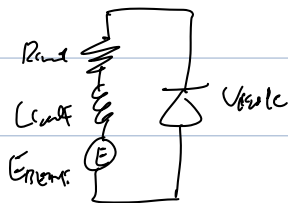


$V_{dc} = R_{load} I_{avg} + L \left(\frac{\Delta I}{t_{on}} \right) + E_{BEMF}$

$\Rightarrow 20 = 0.2 I_{avg} + 0.4m \frac{\Delta I}{375\mu} + 2$

$\Rightarrow I_{avg} = -6.154 \Delta I + 90 \quad (1)$

off state:



$L \left(\frac{\Delta I}{t_{off}} \right) = R_{load} I_{avg} + E_{BEMF} + V_{diode}$

$\Rightarrow 0.4m \cdot \frac{\Delta I}{175\mu} = 0.2 I_{avg} + 2 + 1$

$\Rightarrow I_{avg} = 11.429 \Delta I - 15 \quad (2)$

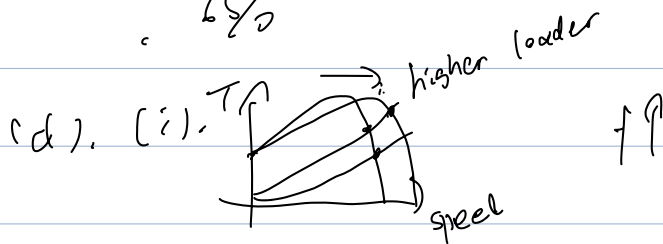
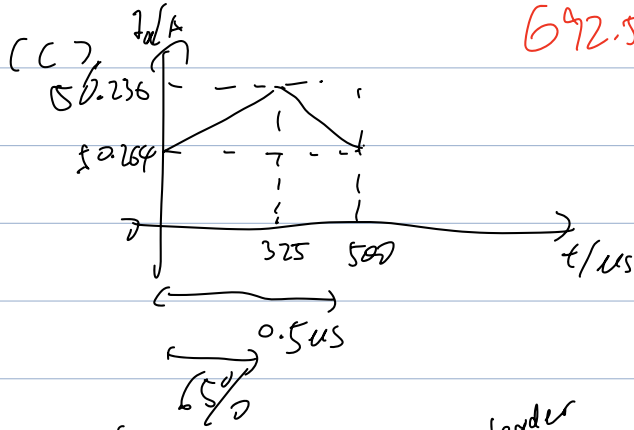
From (1) and (2) $\Rightarrow \begin{cases} I_{avg} = 53.250 A \\ \Delta I = 5.972 A \end{cases}$

Therefore: $I_{max} = I_{avg} + \frac{1}{2} \Delta I = 56.236 A$

$I_{min} = I_{avg} - \frac{1}{2} \Delta I = 50.264 A$

$I_{DC, avg} = I_{avg} = 53.25 A$

(b) $P_{Avg} = I_{Avg} \cdot V_{OL} = \cancel{1.065} \text{ kW}$
 692.51



(ii).

use magnetic with higher B_{max}

or increase the airgap length to increase B_{max} .

so that we can have larger K_e , then
 from $E = K_e \omega$, smaller starting speed.

(e). Due to use of commutator, it is less reliable. The brush/commutator can be substituted by using a drive.

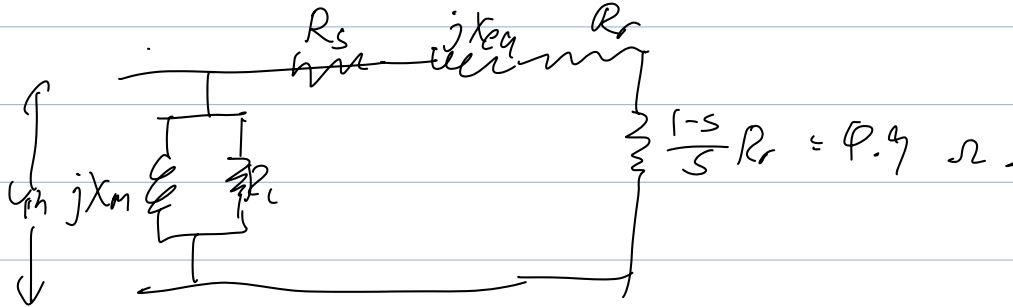
Brushless DC motor

Q2.

(a). From $V_{Lms} = 0.612 \times V_{dc} \times M_a$

$$\Rightarrow M_a = \frac{400}{0.612 \times 800} = 0.817$$

$$c) s = \frac{N_s - R_r}{N_s} = \frac{1500 - 1470}{1500} = 0.02$$



$$Z_T = R_s + R_r + \frac{1-s}{s} R_r + jX_{eq} = 5.124 \angle 5.600^\circ$$

$$|I| = \frac{V_{ph}}{|Z_T|} = \frac{400}{\sqrt{3} \cdot (5.124)} = 45.070 \text{ A}$$

$$P_{mech} = 3 \cdot I^2 \frac{1-s}{s} R_r = 27860.2 \text{ W}$$

$$\Rightarrow T = \frac{P_{mech}}{\omega} = \frac{5484.5}{1470 \times \frac{2\pi}{60}} = \boxed{35.628 \text{ N.m}}$$

$$P_{loss} = 3 I^2 (R_s + R_r) + 3 \cdot \frac{V_{ph}^2}{R_c}$$

$$= 3 \times 45.070^2 \times 0.2 + 3 \left(\frac{400}{\sqrt{3}} \right)^2 \cdot \frac{1}{10}$$

$$= 17218.8 \text{ W}$$

$$\text{Eff} = \frac{P_{mech}}{P_{mech} + P_{loss}} \times 100\% = \boxed{63.43\%}$$

$$c). (i). T_{brake} = -T = \boxed{-35.628 \text{ N.m}}$$

$$E_{860} = \frac{1}{2} C V_{860}^2 = 7396 \text{ J}$$

$$E_{800} = \frac{1}{2} C V_{800}^2 = 6400 \text{ J}$$

$$\Delta E = E_{860} - E_{800} = 996 \text{ J}$$

$$\text{Let } P_{\text{brake}} = 2 P_{\text{mech}} \quad \times 63.43\% \quad \times 98\%$$

$$\text{then } P_{\text{dc}} = (P_{\text{brake}} - P_{\text{mech}}) = 29263.0 \text{ W}$$

$$\Rightarrow t = \frac{\Delta E}{P_{\text{dc}}} = 34.036 \text{ ms}$$

$$\text{C: (i)} \Delta E = \frac{1}{2} C (V_{240}^2 - V_{850}^2) = 507 \text{ J}$$

$$t = \frac{\Delta E}{P_{\text{switch}}} = \frac{1}{25} = 0.04 \text{ s} = 40 \text{ ms}$$

$$t_{\text{rise}} = \frac{\Delta E}{P_{\text{mech}} \times 98\%} = 17.326 \text{ ms}$$

$$t_{\text{fall}} = t - t_{\text{rise}} = 22.674 \text{ ms}$$

$$\Rightarrow P_{\text{dc}} = \frac{\Delta E}{t_{\text{fall}}} = 22360.4 \text{ W} = P_{\text{brake}} - P_{\text{mech}}$$

$$\Rightarrow P_{\text{brake}} = 52220.6 \text{ W}$$

$$R_{\text{brake}} = \frac{V_{\text{brake}}^2}{P_{\text{brake}}} = 13.673 \Omega$$

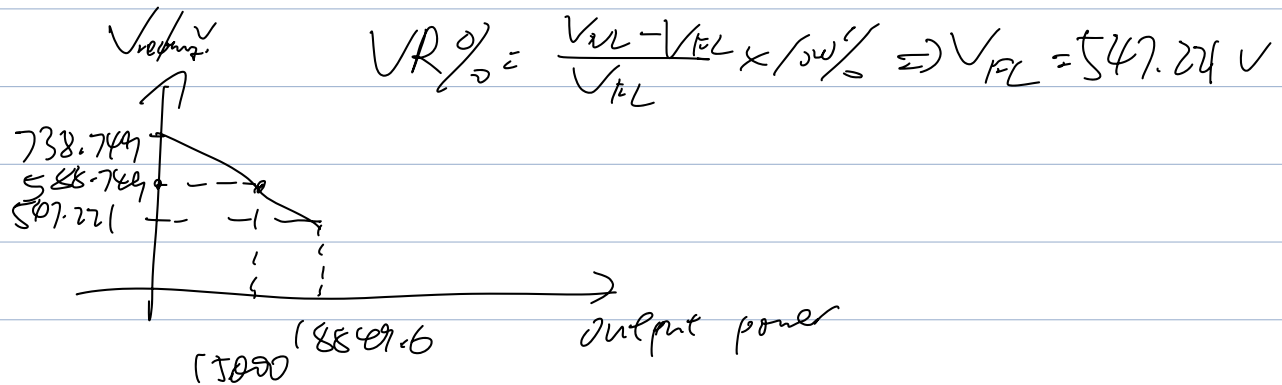
$$\text{Q3, (a), } N_b = \frac{120 f}{p} = \frac{120 \times 60}{6} = 1200 \text{ rpm}$$

$$V_{\text{oc}} E_{\text{BEMF}} = 2.4 \times 1200 \times \frac{2\pi}{60} = 96\pi \approx 301.593 \text{ V}$$

$$\text{when open circuit: } V_{\text{rectifier}} = 12.7 \cdot E_{\text{BEMF}} = 738.749 \text{ V}$$

At max output power point ($\tau = 150 \text{ Nm}$):

$$P_{mech} = \omega \cdot T = 1200 \times \frac{2\pi}{60} \times 150 = 18849.6 \text{ W}$$



$$V_{rectifier} = 738.744 - \frac{738.744 - 547.221}{18849.6} \cdot P_{out}$$

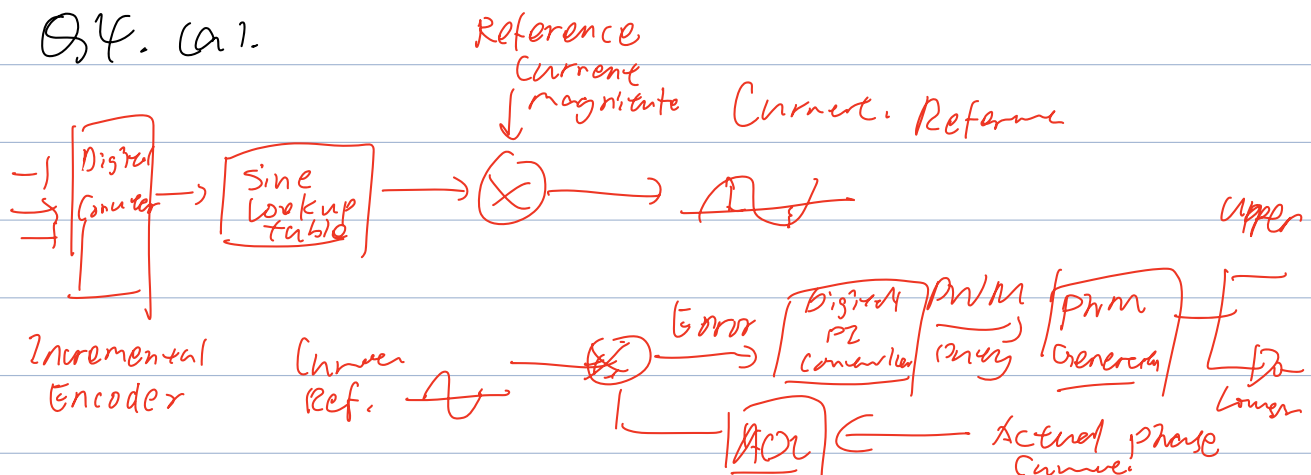
$$= 738.744 - 0.01 P_{out}$$

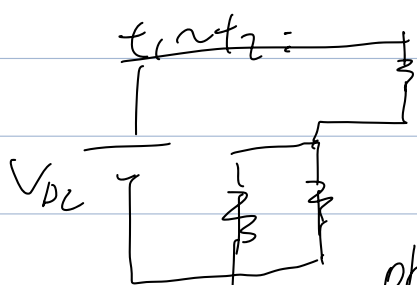
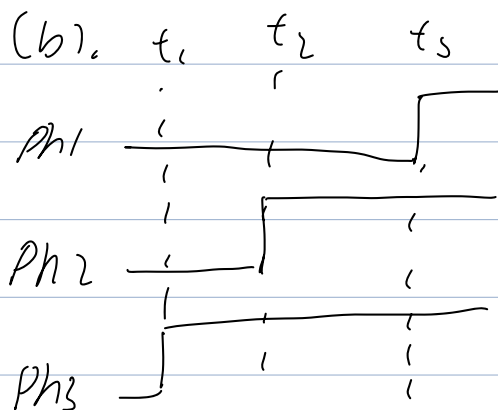
when $P_{out} = 15 \text{ kW} \Rightarrow V_{rectifier} = \underline{588.744 \text{ V}}$

(b) $P_{out} = \frac{V_{out}}{R} \Rightarrow R = \frac{100}{15k} = 6.67 \times 10^{-3} \Omega$

Duty cycle = $\frac{100}{588.744} \times 100\% = 16.985\%$

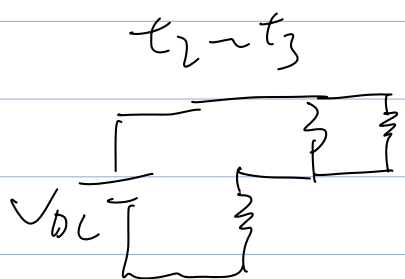
Q4. (a).





$$\text{ph1: } V_{Avg} = 0.3 \times V_{DC} \times \frac{1}{3} + 0.4 \times V_{DC} \times \frac{2}{3}$$

$$= 55V$$



$$\text{ph2: } V_{Avg} = 0.3 \times V_{DC} \times \frac{1}{3} + 0.4 \times V_{DC} \times \frac{1}{3}$$

$$= \cancel{35V} \quad 5V$$

$$\text{ph3: } V_{Avg} = 0.3 \times V_{DC} \times \frac{2}{3} + 0.4 \times V_{DC} \times \frac{1}{3}$$

$$= 50V$$